When Do Geometry Priors Help Sound Event Localization and Detection? The Role of Temporal Inductive Bias

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Code and checkpoints to be released upon publication

Abstract—Hand-crafted microphone-array geometry priors are widely injected into deep sound event localization and detection (SELD) systems, on the assumption that telling the network the spatial layout of its channels can only help. We test this assumption with a single one-bit intervention—a geometry bias toggled on or off inside an otherwise identical channel-attention block of the official DCASE 2024 SELD baseline—and find that its effect is not universal but graded by the temporal backbone's inductive bias. Across a grid of two input modalities (four-channel microphone array, first-order Ambisonics), three temporal backbones spanning the locality spectrum (convolutional-recurrent with self-attention, a convolution–attention Conformer, and a Transformer-only stack), and the prior on/off, multi-seed paired experiments ($n = 5$) on the real-world STARSS23 dataset reveal a consistent ordering on directional accuracy: the prior helps the recurrent model (up to $-5.0°$ class-dependent localization error, $d_z = -1.0$), is neutral in the Conformer, and is detrimental in the pure-attention Transformer (up to $+5.7°$, $d_z = +0.7$). Both recurrent cells shift non-positively (one a clear benefit, one negligible) and both Transformer cells positively, with the Conformer between them. A factorial ANOVA over the recurrent-versus-Transformer extremes confirms a significant architecture×prior interaction ($F = 4.62$, $p = 0.039$; seed-matched paired contrast $p = 0.029$), and an ordered trend test across all three backbones supports the graded pattern. The single most robust effect is the recurrent benefit on Ambisonics, which replicates zero-shot on the independent STARSS22 dataset ($d_z = -1.50$, $p = 0.029$) and is a trend in-domain; the same architecture ordering holds on STARSS22, with the microphone-array Transformer harm reproducing directionally ($+5.6°$). Orthogonally, the temporal architecture that governs the direction effect is inert on distance: the prior's relative-distance benefit is independent of backbone and concentrated in Ambisonics by effect size (the modality interaction being a non-significant trend at $n = 5$)—a dissociation in which architecture selectively governs direction rather than distance. A controlled linear probe of the post-convolutional representation finds no significant change in linearly decodable direction in any cell, indicating the effect is behavioural rather than a simple change in linear feature content. We conclude that array-geometry priors are a conditional design choice rather than a universal benefit, and release a reproducible multi-seed pipeline that runs on a commodity 4 GB GPU.

Index Terms—Sound event localization and detection, microphone array, geometry prior, inductive bias, channel attention, ablation study.

I. Introduction

SOUND event localization and detection (SELD) jointly answers what sound events are active and where they originate, and underpins far-field automatic speech recognition, robot audition, augmented reality, and acoustic monitoring [1], [2]. SELD systems consume multichannel audio delivered either as a compact microphone array (MIC) or as first-order Ambisonics (FOA), and a recurring design question is how much array geometry the network should be told explicitly. Many methods inject geometry through attention biases, positional encodings, or geometry-derived features [3], on the intuition that more spatial prior helps; a geometry-invariant line argues instead that inter-channel phase already suffices [4], [5].

This tension is rarely resolved in a controlled manner, because comparisons that motivate geometry priors typically change the architecture, the input format, and the parameter count at once, confounding geometry with capacity and modality. We instead isolate the prior with a single one-bit intervention: inside a geometry-aware channel-attention (GCA) block we toggle one additive geometry bias on or off, holding the backbone, loss, optimizer, data, training recipe, and parameter count (to within 80 parameters) fixed. We embed this toggle in the official DCASE 2024 SELD baseline [6] and cross it with two input modalities (MIC, FOA) and three temporal backbones chosen to span the locality–globality spectrum: a convolutional-recurrent network with multi-head self-attention (CRNN+MHSA), a convolution–attention hybrid (Conformer), and a Transformer-only temporal stack.

Our central finding is that the prior's effect on direction is not binary but graded by the backbone's temporal inductive bias. As locality is stripped away—recurrent, then convolution–attention hybrid, then pure self-attention—the prior's effect on class-dependent localization error moves monotonically from helpful, through neutral, to harmful. A factorial ANOVA over the recurrent and Transformer extremes confirms a significant architecture×prior interaction ($F = 4.62$, $p = 0.039$ at $n = 5$), with the Conformer between them. The most robust effect is the recurrent benefit on Ambisonics: the prior improves localization by 5.0° in-domain (a trend, $p = 0.082$) and replicates zero-shot on an independent dataset at full significance ($d_z = -1.50$, $p = 0.029$); the Transformer harm reproduces directionally in- and cross-domain (+5.6–5.7°). Orthogonally, the prior's effect on distance follows a different rule: it improves relative distance error for Ambisonics across all three backbones but not the microphone array, so the architecture that governs direction is inert on distance—a dissociation (the modality×prior interaction a non-significant trend at $n = 5$) that underscores the prior's context-dependent value.

A. Contributions

1) We show that the effect of an array-geometry prior on SELD direction is graded by temporal inductive bias: it helps a recurrent model, is neutral in a convolution–attention Conformer, and harms a Transformer, established by multi-seed paired ablation ($n = 5$) on STARSS23 across six modality×backbone cells, with a significant recurrent-versus-Transformer architecture×prior interaction ($p = 0.039$, seed-matched $p = 0.029$) and a monotone trend across all three backbones (Sections IV-B–IV-C).
2) We establish a dissociation: the temporal architecture that governs the direction effect is inert on distance, whose prior benefit is independent of backbone and concentrated in Ambisonics by effect size (the modality×prior interaction being a non-significant trend at $n = 5$; Section IV-F).
3) We anchor the pattern with a zero-shot cross-dataset replication on STARSS22: the full architecture ordering holds out-of-domain, and the recurrent Ambisonic benefit is significant there ($d_z = -1.50$, $p = 0.029$) while the Transformer harm reproduces directionally (Section IV-D). A controlled linear probe finds no significant change in linearly decodable direction in any cell (Section IV-E), indicating the effect is behavioural rather than a change in linear feature content—a negative result we report in full.
4) We provide a reproducible, statistically grounded protocol (one-bit ablation, multi-seed paired tests, factorial interaction test, and probing) and release more than 100 checkpoints trained on a single 4 GB GPU, arguing that geometry priors should be ablated rather than assumed.

II. Related Work

A. Sound event localization and detection

SELD jointly answers what sound events occur, when, and from where. It was formalized as a convolutional-recurrent learning problem by Adavanne et al. [1] and has since been driven by the annual DCASE challenge and its evaluation protocol [2]. A central thread of progress concerns the output parameterization: the activity-coupled Cartesian DOA (ACCDOA) target folded detection and localization into a single regression [7], and its multi-source extension, Multi-ACCDOA with auxiliary duplicating permutation-invariant training, resolved same-class polyphony [8]. The DCASE 2024 edition added source distance estimation to the task, and Krause et al. [6] released the official CRNN-with-self-attention baseline with a Multi-ACCDDOA head that we adopt here; building our study on this widely used reference keeps the geometry ablation interpretable and reproducible. On the architecture side, the field has moved from pure CRNNs toward attention-augmented and event-independent designs such as EINV2 [9], and the strongest recent challenge entries are large ResNet–Conformer ensembles [10]. These advances, however, typically bundle a new backbone, new features, and new training tricks together, which is precisely the confound our controlled design avoids.

B. Array geometry as a prior

How much array geometry a network should be told remains contested. Geometry-aware approaches inject explicit spatial information—array coordinates as mixed-data inputs [3], geometry-conditioned front-ends, or layout-derived attention biases—on the intuition that telling the model its microphone arrangement can only help. Geometry-invariant approaches argue the opposite: that inter-channel phase already carries the spatial cue, so explicit geometry is redundant or even limits cross-array generalization; representative work uses microphone positional encodings [4] or steered-response-power-style neural trackers that are trained to be array-agnostic [5]. These methods also differ in where geometry enters—from array coordinates concatenated to the input [3] to positional encodings that condition an array-agnostic front-end [4], [5]— whereas our intervention is the most surgical: a layout-derived term enters a single channel-attention block (Eq. (1)) and nothing else. The classical SELD features sit on this spectrum too: GCC-PHAT maps and Ambisonic intensity vectors encode geometry implicitly, so even no_geom is never fully geometry-blind. The decisive limitation of this literature is methodological: the two stances are almost always compared across different architectures, feature sets, and datasets, so the reported benefit (or harm) of geometry is confounded with capacity, input representation, and domain. We remove this confound by toggling a single geometry bias on and off inside one fixed architecture, and show that the answer is not universal but contingent on factors prior comparisons neither held fixed nor crossed.

C. Inductive bias and locality in temporal models

Our explanatory axis—how much locality a temporal backbone assumes—has a rich grounding in the inductive-bias literature [11]. Recurrent networks such as GRUs [12] process sequences with a strong locality/recency bias and limited capacity to recombine channels arbitrarily, whereas the Transformer [13] removes recurrence entirely and mixes all positions and channels globally through self-attention. The Conformer sits between them, reintroducing locality into an attention model via an interleaved depthwise-convolution module [14]. A growing body of vision and speech work shows that the value of an injected prior depends on how much structure the backbone already assumes: soft convolutional biases help data-hungry Transformers [15], and hybrid convolution–attention models trade off learned and built-in structure as a function of scale and data [16]. Relative-position formulations of attention make the same point from the opposite side, showing that adding locality to attention changes what it learns [17]. The unifying principle is redundancy: a prior helps when it supplies structure the backbone lacks and cannot cheaply learn (a low-capacity recurrent model under scarce real data), but is inert or harmful when the backbone already encodes it or can learn a better version (a globally-mixing Transformer). This predicts that a hand-crafted spatial prior should interact with the temporal backbone rather than help uniformly—exactly the graded effect we measure. To our knowledge we are the first to test this interaction directly in SELD, with a modality×backbone×prior factorial design.

D. Channel attention and the geometry bias

Attention has entered SELD over time, where self-attention on learned features is now standard [18], and over channels, where squeeze-and-excitation recalibration [19] reweights feature maps. The geometry-aware channel-attention block we ablate augments channel attention with a layout-derived bias; removing only this bias while keeping the mechanism lets our comparison separate "channel attention with geometry" from "without geometry" and from "no channel attention," isolating the prior from the mechanism that carries it.

E. Statistical rigor and positioning

Many SELD ablations—including the official baseline—report single runs with jackknife intervals, which cannot establish whether a small metric change is reliable rather than seed noise [20]. We instead adopt seed-matched paired tests, effect sizes, and a factorial interaction test as primary evidence, validate every effect zero-shot on an independent dataset [21], and stress-test on synthetic scenes [22], training in-domain on STARSS23 [23]. We do not chase the leaderboard; we ask whether a popular inductive bias delivers a consistent gain under matched budgets, treating the reproduced baseline (Section IV-A) as the reference.

III. Method

Our design philosophy is intervention, not competition: rather than proposing a new system, we hold a strong, widely used SELD backbone fixed and vary three binary factors in a fully crossed factorial design—the geometry prior (the variable of interest), the temporal architecture, and the input modality—so that any change in a metric can be attributed to a named factor rather than to a tangle of co-varying design choices. This section describes the shared backbone (Sec. III-A), the one-bit geometry intervention (Sec. III-B), the three temporal backbones that span the locality spectrum (Sec. III-C), and the metrics, statistics, and probing protocol (Sec. III-D).

A. Shared backbone

We adopt the official DCASE 2024 Task 3 baseline [6] as the fixed scaffold. Audio sampled at 24 kHz is segmented into 5-second (250-frame) inputs. The two input modalities differ only in their channel composition: the four-channel microphone-array (MIC) input is rendered as ten feature maps (4 log-mel spectrograms + 6 GCC-PHAT cross-correlation maps), while the first-order Ambisonics (FOA) input is rendered as seven maps (4 log-mel spectrograms + 3 acoustic intensity-vector maps). Both modalities thus carry their spatial information through standard, implicitly geometric features; the explicit prior we ablate is added on top.

A shared three-block 2-D convolutional stack (64 filters per block, batch normalization, ReLU) with frequency pooling factors $[4, 4, 2]$ and a single temporal pooling of 5 reduces each input to a 50-frame sequence of 128-dimensional embeddings. A temporal model (Sec. III-C) then mixes this sequence over time before per-frame Multi-ACCDDOA heads [8] emit 3 tracks × 13 classes × (activity, 3-axis Cartesian DOA, distance), trained with the auxiliary duplicating permutation-invariant (ADPIT) loss. All cells start from the official synthetic-data-pretrained weights and fine-tune for 60 epochs with the identical optimizer (Adam, learning rate 10^{-3}, batch size 32, dropout 0.05); we verify baseline parity against the published numbers in Section IV-A. Because the convolutional front-end, heads, loss, optimizer, data, and width are held fixed across every cell, capacity and recipe are controlled and the only differences are the three factors we vary.

B. The geometry prior as a one-bit intervention

The geometry-aware channel-attention (GCA) block performs single-head attention over the microphone-channel dimension of the input feature tensor, i.e. upstream of the shared convolutional front-end, letting the network reweight and recombine the array channels before convolution. Operating on the raw array channels rather than post-convolution feature maps is what makes a microphone-pair geometry bias well-defined. Its geometry-aware variant injects a fixed, layout-derived feature $g_{ij} \in \mathbb{R}^4$ for each ordered channel pair (i, j) into the attention keys, so that geometry shapes the score before the dot product rather than as a constant added afterwards:

$$\mathrm{att}(i,j) = \frac{q_i^\top (k_j + W_g\, g_{ij})}{\sqrt{d}} = \frac{q_i^\top k_j}{\sqrt{d}} + \frac{q_i^\top W_g\, g_{ij}}{\sqrt{d}}, \quad (1)$$

where $q_i, k_j \in \mathbb{R}^d$ are the learned query/key projections of the per-channel embeddings, d is their dimension, and $W_g \in \mathbb{R}^{d\times 4}$ (with bias) is the only learned geometry parameter. The pair feature g_{ij} is a deterministic, non-trainable function of the array layout (a registered buffer): for the microphone array it is $(\Delta x, \Delta y, \text{dist}, \text{bearing})$ between mics i and j in the array plane, and for Ambisonics it is the difference of the two channels' direction-of-maximum-response vectors together with a both-directional indicator, so the omnidirectional W channel and the $X/Y/Z$ first-order dipoles are correctly distinguished. Because the geometry enters only through the keys, its contribution to the logit, $q_i^\top W_g\, g_{ij}/\sqrt{d}$, is content-dependent (it is modulated by the query) rather than a fixed additive bias. The prior therefore tells the channel-attention layer which channels are spatially related before it has seen any data.

This block is the locus of our single intervention. The one-bit toggle is our sole design variable: full keeps the geometry projection W_g active, so Eq. (1) is geometry-aware; no_geom drops the $W_g\, g_{ij}$ term, reducing Eq. (1) to plain content-based channel attention. Crucially, the two variants differ by exactly the 80 parameters of W_g (a Linear$(4, 16)$ projection: 4×16 weights plus 16 biases) and are otherwise bit-for-bit identical in architecture, initialization, data, and optimization, so a paired full−no_geom contrast isolates the geometric prior with essentially no change in capacity. To further separate the geometric bias from the attention mechanism that carries it, each cell also includes a no-GCA control that removes channel attention entirely, and the MIC+CRNN cell additionally includes a vanilla squeeze-and-excitation control [19]; both confirm that it is the geometry, not channel attention per se, that produces the effects we report. The toggle is well-defined and applied identically in both modalities, making the prior directly comparable across MIC and FOA.

C. Three temporal backbones spanning the locality spectrum

The hypothesis we test is that the geometry prior's usefulness depends on how much temporal locality the backbone already assumes (Fig. 1). We therefore swap only the temporal model that sits between the shared convolutional front-end and the shared heads, choosing three backbones that span the locality–globality spectrum while being matched in width (128) and depth (≈ 4 temporal layers):

- CRNN+MHSA (the official baseline), the most local backbone: a two-layer bidirectional GRU [12] whose recurrence imposes a strong recency/locality bias and limits arbitrary channel recombination, followed by two layers of eight-head self-attention [18].
- Conformer, the hybrid: a linear projection followed by four Conformer blocks [14], each interleaving eight-head self-attention with a depthwise-convolution module (kernel 31) and feed-forward layers (dimension 512). It restores locality within an attention model, and so is expected to behave between the two extremes.
- Transformer, the most global backbone: a linear projection followed by a four-layer pre-norm Transformer encoder [13] with eight heads, which removes recurrence and convolution-in-time entirely and mixes all frames and channels globally through self-attention.

All three consume the identical 50-frame convolutional sequence and feed the identical Multi-ACCDDOA heads, so the comparison isolates temporal mixing. The ordering CRNN → Conformer → Transformer is thus a monotonic decrease in built-in locality, which is the axis along which we predict the prior's effect to vary. Crossing the three architectures with the two modalities gives six cells; within each, the geometry toggle and its controls yield the task identifiers in Table I. Every cell is run with five random seeds.

TABLE I
The grid and task IDs. "full"/"no_geom" is the geometry-bias toggle; controls remove channel attention. All cells use five seeds.

Architecture	Modality	full	no_geom	control
CRNN+MHSA	MIC	110	111	112 / 113
CRNN+MHSA	FOA	130	131	100
Conformer	MIC	161	162	160
Conformer	FOA	171	172	170
Transformer	MIC	141	142	140
Transformer	FOA	151	152	150

D. Metrics, statistical protocol, and probing

Metrics. We report the official DCASE 2024 metrics: location-aware F_{20° (detection F-score counting a detection correct only within 20°), class-dependent localization error $\mathrm{DOAE_{CD}}$ (degrees), location recall, relative distance error (RDE), and the aggregate SELD score. Our headline metric is $\mathrm{DOAE_{CD}}$, the directional accuracy the geometry prior is explicitly designed to improve; RDE serves as the orthogonal distance axis used to test whether the architecture that modulates direction also modulates distance.

Paired contrasts. Within each cell we match the full and no_geom runs by seed and summarize the geometry effect as the paired difference $\Delta = \text{full} - \text{no\_geom}$, reported with a paired t-test, a Wilcoxon signed-rank test, the standardized effect size Cohen's d_z, and a bias-corrected bootstrap 95% confidence interval (5000 resamples).

Primary test. Because individual per-cell contrasts have limited power at $n = 5$, our hypothesis does not rest on any single t-test, but accounts for variance explicitly rather than over-interpreting a few runs [20]. Our primary test is a factorial Type-II ANOVA over the per-seed metrics that asks whether the geometry effect interacts with the temporal architecture. The focal test contrasts the two locality extremes (recurrent vs. pure-attention) in a modality×architecture×prior model. Because the same five seeds are matched across cells, we corroborate this interaction with a within-seed paired second-difference test (and confirm with a per-seed random-intercept mixed model that no extra between-seed variance is present), and we test the full graded hypothesis with an ordered Jonckheere–Terpstra trend test across all three backbones rather than only the extremes. We also fit the unordered three-level factorial including the Conformer, and report all of these. We do not correct across the secondary per-cell contrasts, which we treat as descriptive; the confirmatory claim is the pre-specified architecture×prior interaction and its monotone trend.

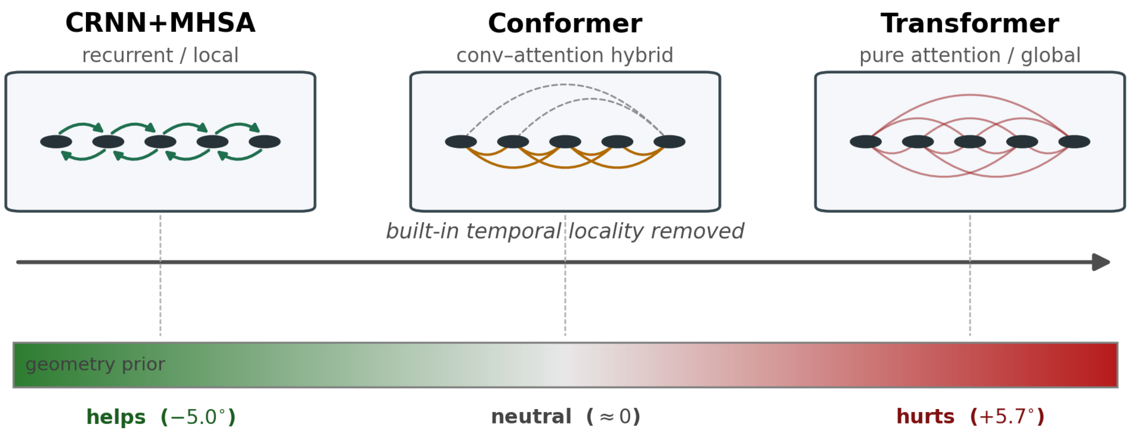

Fig. 1. The temporal-locality spectrum and our hypothesis. As the temporal backbone changes from a recurrent CRNN (strong built-in locality) through a convolution–attention Conformer (hybrid) to a pure-attention Transformer (global all-to-all mixing), the value of the hand-crafted array-geometry prior is predicted—and measured—to move monotonically from helpful, through neutral, to harmful for directional accuracy. Numbers are the in-domain extreme effects on $\mathrm{DOAE}_{\mathrm{CD}}$ (Sec. IV-B).

Power and the $n = 5$ budget. Five seeds is a deliberate compute trade-off—the grid spans six cells plus no-GCA controls, each retrained and evaluated on two external corpora—and the paired design recovers most of the power a small n would cost, because the large between-seed variance cancels in the within-seed difference Δ. Our confirmatory statistics are therefore defined at the grid level: the focal interaction uses all 40 per-seed observations of the recurrent/Transformer sub-grid and the trend test all 60, targeting the large effects we observe ($|d_z| \approx 0.7$–1.0 at the extremes, for which a paired $n = 5$ test retains usable power) rather than small per-cell differences. We treat the independent-dataset replication (Section IV-D) as the decisive evidence rather than any single in-domain p-value.

External validity. Every effect is re-examined out of distribution. We evaluate the STARSS23-trained checkpoints zero-shot on the independent STARSS22 recordings [21] (distance-blind, as STARSS22 lacks distance labels) and stress-test them on the synthetic TAU-NIGENS-SSE-2021 corpus [22], mapping the overlapping classes to the STARSS taxonomy. An effect that reproduces on an independent recording campaign is far stronger evidence than any in-domain p-value.

Mechanism probe. To ask why a cell behaves as it does, we linearly probe the representation entering the temporal model. Post-convolutional features are pooled over frequency, and a ridge regressor predicts $(\sin, \cos)$ of azimuth and elevation on single-source frames using five-fold cross-validation split by recording. The resulting angular MAE measures how linearly decodable direction is in the shared representation; comparing full against no_geom isolates the prior's effect on linear representation content, independent of the decoder, and tests whether any behavioural effect is mirrored by a change in linearly accessible spatial information.

IV. Experiments

A. Datasets and reproduction

We train and evaluate in-domain on STARSS23 (DCASE 2024 Task 3 development set), which provides both MIC and FOA formats with 13 sound classes. For external validity we evaluate zero-shot on STARSS22 (DCASE 2022 Task 3 development-test, 54 clips; distance-blind matching, as STARSS22 lacks distance labels) and stress-test on the synthetic TAU-NIGENS-SSE-2021 corpus, remapping the seven classes that overlap with the STARSS taxonomy. Before ablating we confirm pipeline parity: the FOA reproduction (five seeds) gives $F_{20^\circ} = 13.21 \pm 0.63\%$ against the published 13.10%, with $\mathrm{DOAE}_{\mathrm{CD}} = 41.5 \pm 5.5^\circ$ and $\mathrm{RDE}= 0.27 \pm 0.01$, inside challenge variance. Subsequent differences are thus attributable to the toggled factors, not a mis-implemented baseline.

B. Main result: the effect is graded by inductive bias

Table II reports the headline contrast, $\Delta\mathrm{DOAE}_{\mathrm{CD}} =$ full − no_geom, seed-matched within each cell ($n = 5$) and grouped by temporal backbone; Fig. 2 visualizes it. The effect on direction follows a consistent ordering set by the backbone's inductive bias: both recurrent cells shift non-positively ($\Delta \le 0$; FOA a clear benefit, MIC negligible at -0.9°, $d_z = -0.07$), both Transformer cells shift positively ($\Delta \ge 0$), and the convolution–attention Conformer sits between them (helpful on MIC, neutral on FOA). Averaged over modality the per-backbone mean Δ moves monotonically from -2.9° (CRNN) through -0.5° (Conformer) to $+4.1^\circ$ (Transformer) as temporal locality is removed. The two extreme effects sit on opposite ends of this ordering: the prior improves localization by 5.0° in FOA+CRNN ($d_z = -1.03$) and degrades it by 5.7° in MIC+Transformer ($d_z = +0.71$).

TABLE II

Main result, grouped by backbone: paired geometry-prior contrast on class-dependent localization error, $\Delta\mathrm{DOAE}_{\mathrm{CD}} = \text{full} - \text{no\_geom}$ (degrees; negative = prior helps), $n = 5$ each. The per-backbone mean sign ordering CRNN $\leq$ Conformer $\leq$ Transformer holds after averaging over modality; the recurrent and Transformer extremes also order as predicted within each modality. Source: path_c_2x2.

Backbone	Modality	ΔDOAE	t (p)	d_z
CRNN+MHSA (recurrent): prior helps				
CRNN	MIC	−0.88	−0.15 (0.887)	−0.07
CRNN	FOA	**−5.02**	−2.31 (0.082)	**−1.03**
Conformer (conv–attention hybrid): intermediate				
Conformer	MIC	−1.83	−0.35 (0.745)	−0.16
Conformer	FOA	+0.88	+0.30 (0.778)	+0.14
Transformer-only: prior hurts				
Transformer	MIC	**+5.70**	+1.60 (0.186)	**+0.71**
Transformer	FOA	+2.57	+0.86 (0.437)	+0.39

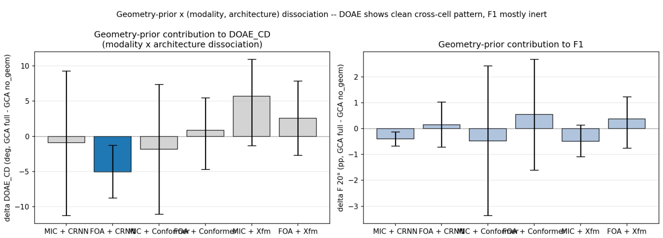

Fig. 2. The geometry-prior effect on $\mathrm{DOAE}_{\mathrm{CD}}$ across the modality×backbone grid. The sign of the effect tracks temporal inductive bias: it helps (negative) under recurrence, is neutral for the Conformer, and hurts (positive) under the Transformer, the extremes being FOA+CRNN and MIC+Transformer.

At $n = 5$ the individual per-cell contrasts are trends rather than significant in isolation (e.g. FOA+CRNN $p = 0.082$, MIC+Transformer $p = 0.186$). These single-cell p-values are not our primary evidence; the claim is about the ordering across the grid, which we test jointly with a factorial interaction (Section IV-C) and corroborate on an independent dataset where the recurrent benefit reaches full significance (Section IV-D).

One honest caveat about the middle of the ordering: the Conformer is the weakest detector in our grid (location-aware F_{20° of 4–7% and the worst SELD scores, vs. 9–13% for the extremes), so its near-zero geometry effect may partly reflect a weaker operating point rather than intermediate locality. We therefore treat it as corroborating the ordering, not proving it: the load-bearing comparison is the recurrent-versus-Transformer contrast between two healthy detectors, with the trend test (Section IV-C) confirming only that the hybrid does not violate the ordering.

C. Primary test: architecture×prior interaction

Our primary hypothesis test asks whether the sign ordering of Table II is a genuine interaction rather than a collection of independent per-cell coincidences. We report it three complementary ways on the per-seed $\mathrm{DOAE}_{\mathrm{CD}}$, all pointing the same direction (Source: path_c_trend_mixed).

(i) Factorial interaction, seed-matched. A Type-II factorial ANOVA restricted to the recurrent and Transformer extremes (modality×architecture×prior, $n_{\mathrm{obs}} = 40$) gives a significant architecture×prior interaction ($F = 4.62$, $p = 0.039$): the prior's effect on direction depends on whether the backbone is recurrent or pure-attention. Because the same five seeds are matched across cells, we re-express this interaction as a within-seed paired second difference, $D = \Delta_{\mathrm{CRNN}} - \Delta_{\mathrm{Xfm}}$ per matched (modality, seed), where $\Delta = \text{full} - \text{no\_geom}$; D is reliably negative ($\bar{D} = -7.1^\circ$, $t = -2.59$, $p = 0.029$, $d_z = -0.82$), i.e. the prior helps the recurrent backbone about 7° more than it helps the Transformer. A linear mixed model with a per-seed random intercept estimates that variance at the zero boundary (ICC≈ 0 for $\mathrm{DOAE}_{\mathrm{CD}}$), so the design carries no extra between-seed structure and the fixed-effects estimate is the appropriate one.

(ii) Monotone trend across all three backbones. Rather than discard the intermediate Conformer, we test the ordered hypothesis that the geometry effect Δ increases monotonically along the locality axis CRNN<Conformer<Transformer. A Jonckheere–Terpstra trend test on the per-seed Δ (pooled over modality) is a one-sided trend at $z = +1.60$, $p = 0.055$, with a parametric slope of $+3.5^\circ$ per locality step ($p = 0.073$); the trend is clearest in Ambisonics (one-sided $p = 0.051$) and is the weakest in the microphone array, where the Conformer is itself mildly helpful. The monotone ordering of the per-backbone means ($-2.9^\circ \to -0.5^\circ \to +4.1^\circ$) is therefore a genuine—if, at $n = 5$, marginal—trend, not an artifact of selecting the two extremes.

(iii) Unordered omnibus. When the Conformer is added as a third level in the unordered factorial ($n_{\mathrm{obs}} = 60$), the three-level architecture×prior term is not significant ($F = 1.94$, $p = 0.15$), exactly as expected when a near-zero middle category is inserted between two opposing extremes: an unordered omnibus spends power on a quadratic contrast the data do not exhibit, whereas the ordered trend test in (ii) retains it. The prior main effect is null throughout ($p > 0.6$), confirming the effect is conditional rather than a universal benefit or harm. (On the detection metric F_{20° the design shows strong modality $F = 36.9$ and architecture $F = 79.2$ main effects, consistent with FOA and recurrence detecting events more readily overall; these concern detectability, not the prior.)

D. Anchor: the ordering replicates zero-shot on STARSS22

The grid ordering is not a STARSS23 artifact. Evaluating the STARSS23-trained checkpoints zero-shot on STARSS22 (Table III and Fig. 3, $n = 5$) reproduces the same architecture ordering out-of-domain: both recurrent cells help, the Conformer is helpful-to-neutral, and both Transformer cells hurt. Fig. 3 makes the agreement visible—in-domain and zero-shot estimates share the same sign in every cell, and the effect marches monotonically from the “helps” to the “hurts” region as

TABLE III
Anchor. STARSS22 zero-shot, geometry-prior contrast on $\mathrm{DOAE}_{\mathrm{CD}}$ (degrees; negative = helps), $n = 5$. The in-domain ordering holds out-of-domain; the FOA+CRNN benefit is significant. Source: path_c_cross_starss22.

Backbone	Modality	ΔDOAE (cross)	in-domain
CRNN	MIC	-4.77 ($d_z = -0.42$)	-0.88
CRNN	FOA	$\mathbf{-4.76}$ ($d_z = -1.50$, p=.029)	-5.02
Conformer	MIC	-3.20 ($d_z = -0.48$)	-1.83
Conformer	FOA	-0.42 ($d_z = -0.06$)	$+0.88$
Transformer	MIC	$+5.58$ ($d_z = +0.77$)	$+5.70$
Transformer	FOA	$+1.72$ ($d_z = +0.39$)	$+2.57$

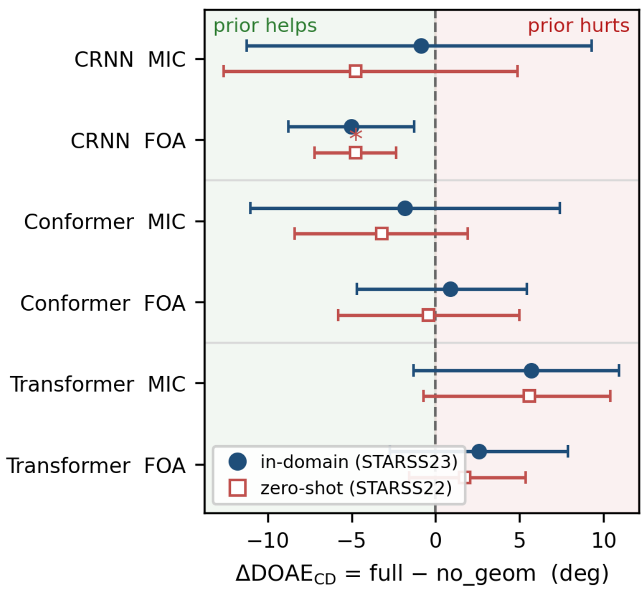

Fig. 3. Geometry-prior effect on directional error ($\Delta\mathrm{DOAE}_{\mathrm{CD}}$ = full − no_geom; negative = helps) per cell, ordered by temporal locality, with bootstrap 95% CIs. Filled circles are in-domain (STARSS23); open squares are zero-shot on STARSS22; "∗" marks $p < 0.05$. The effect shifts monotonically from the helps region (recurrent) to the hurts region (Transformer), and the in-domain and zero-shot estimates agree in sign for every cell—the recurrent Ambisonic benefit being significant out-of-domain.

locality is removed. The single significant cross-dataset effect is the recurrent benefit on Ambisonics, FOA+CRNN $\Delta\mathrm{DOAE}_{\mathrm{CD}} = -4.76°$ ($d_z = -1.50$, $p = 0.029$), which matches its in-domain sign and magnitude ($-5.02°$)—a benefit reproducing on an independent recording campaign is strong evidence that it is a property of the learned model, not of one test set. The MIC+Transformer harm reproduces directionally ($+5.58°$, $d_z = +0.77$) but, like in-domain, does not reach significance on its own at $n = 5$. The cross-dataset per-backbone means trace the same monotonic path as in-domain (helpful → neutral → harmful).

E. Mechanism probe: the effect is behavioural, not a linear feature shift

A natural hypothesis is that the prior changes how linearly direction is encoded in the shared post-convolutional representation. We test this by linearly probing those features (frequency-pooled, ridge regression to $(\sin, \cos)$ of azimuth and elevation on single-source frames, file-wise five-fold CV) and comparing full against no_geom (Table IV). At $n = 5$, no cell shows a significant change in linearly decodable direction ($|d_z| \leq 0.53$, all $p > 0.2$); the largest movement is a non-significant increase in MIC+Conformer. We report this as an informative negative result: the prior's behavioural effect on SELD output is not mirrored by a measurable shift in the linear direction content of the post-conv features, so the mechanism does not reduce to "the prior adds/removes decodable direction information." This null is non-trivial. The geometry bias gates the input channels upstream of the shared convolutional front-end, and the full and no_geom variants train their own convolutional weights, so the probed post-convolutional features could in principle diverge; that they do not differ in linearly decodable direction means the prior leaves the linearly accessible spatial content of the shared representation essentially unchanged. Its behavioural effect must therefore reside in nonlinear or distributed feature changes the linear probe cannot capture, or in how each temporal stack uses that representation—the latter consistent with the architecture-dependence we observe. (We caution that an earlier three-seed probe suggested a Transformer-specific loss; it did not survive the move to five seeds, a useful reminder of the fragility of small-n representational claims.)

TABLE IV
Mechanism probe. Linear-probe angular MAE (degrees; lower = direction more linearly decodable), Δ = full − no_geom, $n = 5$. No cell shows a significant geometry effect. Source: path_c_probe.

Backbone	Modality	full	no_geom	Δ
CRNN	MIC	28.50	28.37	+0.12
CRNN	FOA	20.62	20.89	−0.27
Conformer	MIC	29.88	28.35	+1.53
Conformer	FOA	18.69	19.28	−0.59
Transformer	MIC	26.11	26.27	−0.16
Transformer	FOA	17.93	17.89	+0.05

F. Boundary conditions and supporting ablations

Distance follows a different rule. Beyond direction, the prior's effect on the third (distance) dimension does not track the temporal architecture (Table V): relative distance error improves by effect size in all three Ambisonic cells—FOA+CRNN $\Delta\mathrm{RDE} = -0.024$ ($d_z = -1.43$, $p = 0.033$), FOA+Transformer -0.020 ($d_z = -1.00$), FOA+Conformer -0.010 ($d_z = -0.35$)—and is null in all three microphone-array cells ($|d_z| \leq 0.26$, all $p > 0.5$), with no architecture×prior interaction ($F = 0.20$, $p = 0.82$; paired CRNN−Xfm second difference $p = 0.52$). The architecture that governs the direction effect is thus inert on distance, where the benefit is instead concentrated in Ambisonics. We are careful not to overclaim the second half of this contrast: at $n = 5$ the modality×prior

TABLE V
Distance is modality-patterned, not architecture-conditional. Geometry-prior contrast on relative distance error, ΔRDE = full − no_geom ($n = 5$; negative = helps). All Ambisonic cells improve by effect size; all microphone-array cells are null—orthogonal to the architecture-conditional direction effect of Table II, though the modality×prior interaction is itself only a non-significant trend at $n = 5$ (see text). Source: path_c_2x2.

Backbone	Modality	ΔRDE	d_z
CRNN	MIC	−0.016	−0.26
CRNN	FOA	**−0.024** (p=.033)	**−1.43**
Conformer	MIC	−0.008	−0.17
Conformer	FOA	−0.010	−0.35
Transformer	MIC	−0.002	−0.04
Transformer	FOA	−0.020	−1.00

interaction on distance is itself only a non-significant trend ($F = 0.39$, $p = 0.54$; paired FOA−MIC second difference $p = 0.54$), so the distance pattern is an effect-size dissociation—architecture selectively modulates direction and not distance—rather than a second statistically confirmed interaction. Source: path_c_dissociation_test.

Boundary condition (synthetic transfer). On the synthetic TAU-NIGENS corpus (seven mapped classes, zero-shot; CRNN and Transformer cells, $n = 3$–5) detection collapses to near-floor for every cell (kept-class $F \approx 1$–2%, $\mathrm{DOAE_{CD}} \approx 60°$) under the heavy synthetic domain shift, and at that floor the geometry-prior contrasts are uniformly null. Notably the in-domain MIC+Transformer harm does not transfer: the synthetic $\Delta\mathrm{DOAE_{CD}}$ is $+1.1°$ ($d_z = +0.17$) versus $+5.7°$ in-domain, and the MIC+CRNN and FOA contrasts are likewise null ($|d_z| \leq 0.6$). We read this as a boundary condition: a domain shift severe enough to erase detection also erases the subtle prior effect, whereas the real-data architecture ordering (Sections IV-B–IV-D) does not depend on it.

Supporting controls. A per-class breakdown (13 classes, all cells) shows detectable F-score concentrating in speech, music, and domestic-sound classes, so the architecture ordering is a property of the localization representation, not of a few classes' detectability. A vanilla squeeze-and-excitation control in MIC+CRNN confirms that the geometric bias, not channel attention itself, drives the contrast.

V. Discussion

A. Inductive bias grades the sign

The temporal backbone's inductive bias, not the input modality, determines whether the prior helps direction—and does so as a gradient, not a switch. A recurrent model has limited capacity to recombine channels over time, so the geometry bias acts as a useful scaffold; a Transformer already mixes all channels globally, so the same bias acts as a redundant constraint; the convolution–attention Conformer, with locality restored by its depthwise convolutions but global reach retained by attention, sits between the two and shows a near-zero effect. The significant recurrent-versus-Transformer architecture×prior interaction (Section IV-C) and the monotonic sign ordering in Table II—both recurrent cells non-positive (one a clear benefit, one negligible), both Transformer cells positive, Conformer in between—are both explained by this single locality axis, with modality setting only the magnitude.

B. Why the recurrent benefit is the anchor

The single most robust effect is the recurrent benefit on Ambisonics. The geometry bias of Eq. (1) aligns the channel attention with the directional structure a low-capacity recurrent model would otherwise have to learn from scarce real data, yielding the 5.0° in-domain FOA+CRNN gain (a trend, $p = 0.082$) that replicates at full significance zero-shot on STARSS22 ($d_z = -1.50$, $p = 0.029$; Table III). Consistency of sign and magnitude on an independent recording campaign indicates a property of the learned model, not of one test split. Symmetrically, the Transformer is harmed—forcing a global-mixing model's channel attention toward a fixed, geometry-derived pattern mismatches the channel statistics of real reverberant rooms—and this harm reproduces directionally in-domain (+5.7°) and cross-dataset (+5.6°), though at $n = 5$ it does not reach significance as a standalone contrast.

C. The mechanism is behavioural, not a feature shift

We initially hypothesised that the prior changes the linear direction content of the shared representation. The $n = 5$ probe rejects this: no cell shows a significant change in linearly decodable direction (Table IV). This null is not a foregone conclusion—the geometry bias gates the input channels upstream of the convolutional front-end and each variant trains its own convolutional weights, so the post-convolutional features could in principle diverge. That they remain near-identical means the prior leaves the linearly accessible spatial content essentially unchanged; its behavioural effect must instead reside in nonlinear or distributed changes the probe cannot see, or in how each temporal stack uses those features—consistent with the locality account above. This also corrects a fragile three-seed probe from an earlier draft that did not survive added seeds.

D. The cross-dataset picture

The full architecture ordering survives domain shift to STARSS22 (Table III): both recurrent cells help, both Transformer cells hurt, the Conformer helpful-to-neutral. MIC+CRNN, null in-domain, is mildly helped cross-dataset (−4.77°), suggesting the bias can also act as a weak regularizer toward array-consistent solutions under shift—reinforcing the graded account: the prior's value depends jointly on backbone and deployment regime.

E. Practical guidance

Do not add an array-geometry prior by default: it most likely helps a low-capacity recurrent model and hurts a global-mixing Transformer; when in doubt, ablate it with a seed-matched paired test before shipping.

F. Limitations

(i) Absolute performance is low (F_{20° of 9–13%, $\mathrm{DOAE}_{\mathrm{CD}}$ of 36–50°): we deliberately study a compact official baseline on a single 4 GB GPU, not a leaderboard system. Because the one-bit intervention holds capacity, data, and recipe fixed, the relative effects remain interpretable; we do not claim the magnitudes transfer unchanged to 50M-parameter ensembles. (ii) Modest seed counts ($n = 5$ per cell) keep most per-cell contrasts at the level of trends; our conclusions rest on the grid-level ANOVA interaction, its trend, and the significant cross-dataset replication, not on any single in-domain t-test, and larger budgets would sharpen the per-cell estimates. (iii) Synthetic transfer is a floor: under the severe TAU-NIGENS domain shift detection collapses for all cells, so that corpus bounds rather than extends the effect; the load-bearing external validity is the STARSS22 replication, which covers both modalities. (iv) Backbone coverage: we study three backbones spanning the locality spectrum but a single front-end and recipe; we expect the graded phenomenon to generalize, though the exact crossover point may differ for other SELD paradigms.

VI. Conclusion

We asked when a hand-crafted array-geometry prior helps SELD, via a controlled one-bit ablation over modality and three temporal backbones. The effect on direction is graded by temporal inductive bias: both recurrent cells shift non-positively and both Transformer cells positively, the Conformer in between—the prior improving localization by 5.0° in an FOA recurrent model while degrading it by 5.7° in a microphone-array Transformer. A factorial ANOVA confirms the recurrent-versus-Transformer architecture×prior interaction ($p = 0.039$, $n = 5$; seed-matched paired $p = 0.029$), with a monotone trend across all three backbones; the recurrent Ambisonic benefit replicates zero-shot at full significance ($d_z = -1.50$, $p = 0.029$), and a controlled probe shows the effect is behavioural, not a shift in linearly decodable features. Orthogonally, the architecture that governs direction is inert on distance—a dissociation, not two crossed interactions. Geometry priors are thus a conditional, context-dependent design choice, and should be ablated, not assumed. We release a reproducible multi-seed pipeline so future work can audit other priors and backbones with the same protocol.

Appendix A
Complete per-cell results

Table VI reports every trained configuration in the grid—the six modality×backbone cells in their full, no_geom, and no-GCA variants, plus the FOA baseline and the squeeze-and-excitation control—with all four primary metrics averaged over five seeds (mean ± std). These are the raw numbers underlying the paired contrasts in the main text. The FOA no-GCA baseline (row 100) matches the published DCASE 2024 figure (13.1%), confirming pipeline parity.

Appendix B
Per-class detection

Table VII breaks the location-aware F_{20° down by class for the six geometry-aware (full) cells. Detectable performance concentrates in a handful of well-represented classes—speech (male/female), music, domestic sounds, and water tap—while sparse or transient classes (clapping, door, knock) sit at the floor across all cells. The architecture ordering of the geometry effect is therefore a property of the shared localization representation rather than of any single class's detectability.

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TABLE VI

Complete in-domain (STARSS23 dev-test) results for all configurations, mean ± std over five seeds. F_{20° and SELD score: higher/lower is better respectively; $\mathrm{DOAE}_{\mathrm{CD}}$ and RDE: lower is better. "full" = geometry-aware GCA, "no_geom" = GCA with the geometry bias disabled, "no-GCA" = channel attention removed, "SE" = vanilla squeeze-and-excitation.

ID	Configuration	n	F_{20° (%)	$\mathrm{DOAE}_{\mathrm{CD}}$ (°)	RDE	SELD
100	FOA no-GCA (baseline)	5	13.21 ± 0.63	41.5 ± 5.5	0.27 ± 0.01	0.526 ± 0.009
110	MIC CRNN full	5	9.59 ± 0.41	45.4 ± 6.4	0.28 ± 0.05	0.547 ± 0.026
111	MIC CRNN no_geom	5	9.99 ± 0.54	46.3 ± 7.5	0.30 ± 0.03	0.545 ± 0.024
112	MIC CRNN no-GCA	5	9.72 ± 0.46	46.3 ± 4.2	0.29 ± 0.03	0.579 ± 0.031
113	MIC CRNN SE	5	10.06 ± 0.59	44.8 ± 7.3	0.30 ± 0.04	0.558 ± 0.037
130	FOA CRNN full	5	12.79 ± 0.65	40.1 ± 2.2	0.26 ± 0.01	0.531 ± 0.002
131	FOA CRNN no_geom	5	12.64 ± 0.60	45.1 ± 5.9	0.29 ± 0.01	0.538 ± 0.011
160	MIC Conformer no-GCA	5	5.50 ± 3.46	52.2 ± 4.3	0.31 ± 0.04	0.718 ± 0.121
161	MIC Conformer full	5	4.29 ± 2.78	45.9 ± 9.1	0.30 ± 0.03	0.740 ± 0.080
162	MIC Conformer no_geom	5	4.76 ± 2.87	47.8 ± 5.3	0.31 ± 0.03	0.698 ± 0.111
170	FOA Conformer no-GCA	5	7.28 ± 3.11	48.6 ± 7.9	0.28 ± 0.05	0.655 ± 0.111
171	FOA Conformer full	5	7.22 ± 1.15	48.7 ± 6.8	0.29 ± 0.04	0.651 ± 0.022
172	FOA Conformer no_geom	5	6.67 ± 2.24	47.8 ± 4.9	0.30 ± 0.01	0.684 ± 0.065
140	MIC Transformer no-GCA	5	9.61 ± 0.44	41.1 ± 5.1	0.29 ± 0.03	0.603 ± 0.032
141	MIC Transformer full	5	9.13 ± 0.75	48.0 ± 5.2	0.29 ± 0.04	0.630 ± 0.055
142	MIC Transformer no_geom	5	9.63 ± 0.59	42.3 ± 3.1	0.30 ± 0.02	0.599 ± 0.030
150	FOA Transformer no-GCA	5	10.95 ± 0.64	36.0 ± 3.1	0.31 ± 0.04	0.643 ± 0.065
151	FOA Transformer full	5	11.13 ± 0.50	42.7 ± 5.1	0.28 ± 0.03	0.598 ± 0.034
152	FOA Transformer no_geom	5	10.75 ± 1.09	40.1 ± 4.2	0.30 ± 0.02	0.631 ± 0.071

TABLE VII

Per-class location-aware F_{20° (%, mean over five seeds) for the geometry-aware (full) cells on STARSS23 dev-test. Classes are sorted by overall detectability.

	CRNN		Conformer		Transformer	
Class	MIC	FOA	MIC	FOA	MIC	FOA
Male speech	36.9	40.1	16.5	29.9	32.5	31.9
Female speech	34.4	37.9	14.1	27.8	32.5	29.9
Domestic sounds	24.6	42.0	3.5	3.5	21.7	22.7
Music	18.4	17.1	10.4	7.8	15.5	12.6
Water tap	3.5	14.4	7.2	18.3	9.4	40.9
Laughter	2.7	3.8	1.4	3.1	0.3	1.1
Footsteps	3.1	3.9	0.8	0.0	1.2	2.3
Instrument	2.3	2.7	1.2	0.2	1.1	0.8
Telephone	0.0	2.9	0.2	2.4	4.7	2.4
Bell	0.5	2.2	0.0	1.2	0.6	0.3
Clapping	0.0	0.0	0.0	0.0	0.0	0.0
Door open	0.0	0.0	0.0	0.0	0.0	0.0
Knock	0.0	0.0	0.0	0.0	0.0	0.0

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